

Praise with a Raise: Community-based Gameful Research for Supporting Women and Children Experiencing Homelessness

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Abstract

We present a community-based research collaboration with *Marillac Place*, a non-profit organization supporting women and children experiencing homelessness, to co-design a gameful digital tool for their Praise with a Raise (PWAR) program. Through iterative co-design sessions with staff and interdisciplinary collaboration, we designed app prototypes aimed at enhancing resident engagement by incorporating gamified elements such as badges, progress tracking, and personalized tasks. Despite the co-design efforts, funding limitations prevented the full implementation of the app, reflecting broader challenges in translating academic research into deployable solutions for community-based organizations. We discuss the importance of robust funding mechanisms to support all phases of community-based projects and advocate for stronger academic-community partnerships to ensure real-world impact in addressing societal challenges.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **User studies**.

Keywords

community-based research, gamification, case study

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1 Introduction

Games research often seeks to provide novel ways of motivating behaviour change [4]. It has done so through integration of theories for human motivation like Self-Determination Theory [15], to create engaging supports for people to improve aspects of their life like their health [2, 5], finances [9] or education [14]. However, to date, the HCI community has largely focused on theory development, testing, and academic prototypes — not community-based research in which gameful platforms are deployed and studied in practice.

We were approached by a local non-profit organization, called *Marillac Place*, to develop a gameful platform to support their residents. Marillac Place provides a safe home for women and girls experiencing homelessness and who are pregnant, or who have a child in their care, or who are attempting to regain care of their child(ren). Residents at Marillac Place participate in a program called *Praise with a Raise* (PWAR) that is designed to help them develop key supports like a housing plan and parenting skills. However, compliance with the program did not always meet Marillac Place's expectations, and they hoped that a gameful app might be a means of improving participation and program outcomes.

In this work, we describe our requirements gathering and design work with Marillac Place — including a series of group interviews and feedback sessions with staff, and the development of wireframe prototypes for a gameful platform to support the PWAR program. Our developed prototypes include interfaces for both residents and staff, and employ gameful techniques like points, personalization, and rewards to motivate residents. We reflect on our experience and discuss how our research provides a valuable case study for real-world implications and constraints between academic and community-based initiatives.

2 Related Work

Gamification offers a powerful approach for behaviour change by integrating nudging theory into gameful design. Nudging theory, rooted in behavioural economics, focuses on shaping environments that guide decision-making subtly, encouraging users to make positive choices without removing alternatives [3]. In gamified systems, nudges often take the form of feedback loops, progress tracking, and

rewards like badges or points [17, 19]. Mamede et al. [11] found that combining nudges (e.g., motivational messages) with gamified elements (e.g., progress tracking) was effective in reducing sedentary behaviour by encouraging walking breaks. By embedding nudging principles into game mechanics—such as streaks and milestones, gamification can encourage incremental behaviour change. Applications that incorporate immersive themes and avatars can nudge users toward sustained engagement by creating environments that are both purposeful and enjoyable [8]. Techniques such as tailored reminders, progress visualizations, and customization features allow users to align game elements with their personal goals [12]. These approaches are particularly relevant in community-based contexts, where systems must be sensitive to the unique challenges faced by vulnerable populations. For example, ethical gamification can empower women and children in shelters by providing tools to track goals, access resources, and celebrate achievements, fostering a sense of agency and self-efficacy [10].

2.1 HCI & Community-Based Research

Human-Computer Interaction (HCI) research increasingly integrates community-based approaches to address the specific needs of non-profits and marginalized communities. Participatory methods in HCI prioritize co-design with end-users, ensuring that digital solutions reflect lived experiences and address real-world challenges. This approach is particularly important in non-profit contexts, where tools often serve vulnerable populations, such as women and children in community shelters, who face emotional, social, and logistical barriers to accessing support [10].

Community-based research in HCI emphasizes iterative and inclusive design processes. By involving users in the development cycle, researchers can ensure that digital interventions are both effective and culturally relevant. For example, gamified tools co-designed with community members can provide tailored nudges, such as positive reinforcement and goal-setting prompts, that align with users' needs and foster meaningful engagement. These tools not only enhance usability but also promote equity and inclusivity by addressing systemic barriers to access and participation.

2.2 Ethical Frameworks for Community-Based Research

Ethical frameworks are critical in community-based research, particularly when working with vulnerable populations [18]. Designing for marginalized groups requires empathy and a commitment to upholding dignity, autonomy, and inclusivity. Kilmer et al. [10] stress the importance of creating systems that respect individual differences and avoid a one-size-fits-all approach. Ethical considerations are especially crucial in gamification, where poorly designed rewards or nudges could unintentionally exploit or marginalize users. Ethical gamification incorporates principles of user-centered design to balance motivational strategies with respect for user agency. For instance, systems designed for women and children in shelters must avoid overemphasizing extrinsic rewards and instead focus on fostering intrinsic motivation through personalization, goal-setting, and positive reinforcement [12]. By embedding ethical principles into the design process, researchers can create tools

that empower users, facilitate access to resources, and promote long-term behavior change in equitable and inclusive ways.

3 Research Context

Marillac Place is a small, not-for-profit organization that supports families in the Waterloo Region. It helps people aged 16 and older who are pregnant, have children under 2 years old, or are trying to regain custody of their children. The organization offers key health services—such as housing, parenting and life skills programs, and children's supports—with about 10 full-time employees and an annual budget of approximately \$600,000 CAD at the time of our study. Marillac Place tackles social determinants of health, including housing, income, social connections, and education, using trauma-informed care while promoting inter-generational health through parenting and children's services [1, 7]. It approached the research team to help develop a novel, gameful version of *Praise with a Raise* (PWAR).

3.1 Motivation/Problem/Opportunity

Residents can live at Marillac Place for up to 12 months while participating in the PWAR program, which builds essential life, parenting, and social skills for independent living. Participants earn external rewards—such as credits for toiletries, toys, or household items—and receive oversight through house reviews, weekly staff meetings, life skills sessions, and required chores. The program is managed in three steps: first, staff track progress on paper; second, they enter the data into Excel spreadsheets; and third, the data is integrated into Power BI to review trends. This manual process delays residents' access to information. Although effective initially, the system needs to better engage participants to improve outcomes. Our team designed a gamified digital app prototype to support behavior change, enhance access to data and goals for residents, and provide staff with improved monitoring and trend analysis [13, 16].

3.2 Objectives/Goals

The long-term goal of this research is to derive requirements for, conceptualize and design a novel digital platform to improve health behaviour change through the PWAR program. To accomplish this goal, the team has set out three short-term research activities:

Activity 1: Requirements gathering with Marillac Place staff and residents

Activity 2: Low-Fidelity Prototyping

Activity 3: Joint Co-Design Workshops

After developing rapport with Marillac Place, understanding of what Marillac Place was hoping to obtain as project deliverables and obtaining ethics clearance (ORE #44421), the research team began the process of requirements gathering through two main phases: document review and interviews/focus groups. Participants were eligible to participate in the interviews/focus groups as long as they were Marillac Place staff.

3.3 Requirements Gathering

To understand the scope of the existing PWAR program, the research team requested documentation related to the PWAR program

and conducted a document review on 13 documents sent over by Marillac Place. These documents included general program information, program goals and expected outcomes; existing program recording sheets; resident program checklists (i.e., for chores); staff safety checklists (i.e., room checks to ensure health and safety); resident housing plan; baby check-ins; and resident's program goals.

The research team was invited to join Marillac Place's internal "Innovation Meetings" regularly held among Marillac Place staff to brainstorm improvements/changes to their program delivery. During the Innovation Meeting, the research team learned about staff roles and responsibilities, how the program typically runs on a day-to-day basis, as well as current successes and challenges with the PWAR program. The research team returned to another Innovation Meeting to present to the Marillac Place staff to help staff gain an understanding of the research project and a brief introduction to how gamification and gameful design might fit into the PWAR program. We then asked Marillac Place staff to provide the team with a list of initial requirements and through consultation with Marillac Place, these requirements were grouped into "Must-haves", "Should-haves", and "Could-haves."

3.4 Co-Design Process

We interviewed a total of 6 Marillac Place staff through a series of 3 semi-structured focus groups and individual interviews and prototype development phases.

3.4.1 Initial Focus Groups. The research team facilitated requirements gathering (Activity 1) through a series of semi-structured interviews with Marillac Place front-line staff and Marillac Place administration staff, as well as a review of existing documentation relating to the current implementation of the PWAR program (i.e., task sign-up sheets), and notes/recordings/presentations from the internal "Innovation Meetings" regularly held among Marillac Place staff to brainstorm improvements/changes to their program delivery. We met with six (6) Marillac Place staff during their Innovation Meetings to understand how the PWAR program was administered and to understand the initial needs of the Marillac Place staff. Initial questions included:

- (1) What do you look for in a digital application or game?
- (2) What games do you play and what features in these games do you find compelling?
- (3) What features in the current "PWAR" program would you like (or not like) to see highlighted in the digital app?
- (4) What are features that would encourage you to use the app more (for example, regular notifications/reminders, daily challenges (mini-games), etc.)?

3.4.2 Low-Fidelity Prototype Probes. The requirements provided by Marillac Place staff were then used to develop a series of low-fidelity design prototype probes. These probes were intended to be provocative, aspirational, and to explore potential research directions rather than those that were immediately feasible. They were low fidelity prototypes; that is, they only provided an abstract sense of the final interface but enabled rapid exploration and discussion of possible design directions. While the research team worked on the low fidelity prototypes, we communicated several times with the Third Author to ask questions and receive clarification

on the PWAR program and the day-to-day routine of the house. The Third Author's knowledge of the Marillac Place house, PWAR program and staff responsibilities and routines helped shape our design process and ultimately, the final 6 low-fidelity prototypes.

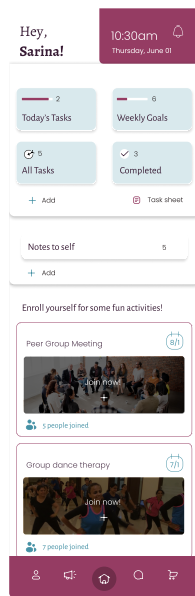
3.4.3 Follow-Up Focus Groups. After the low-fidelity design probes were completed, the research team interviewed three (3) Marillac Place staff to receive feedback and suggestions for change. In these subsequent semi-structured focus groups, we asked Marillac Place staff what they liked about the current prototypes, considerations around accessibility and ease of use, as well as what changes they might like to see in subsequent iterations of the prototypes.

3.4.4 Medium-Fidelity Prototypes. Using the feedback from the Marillac Place team, the research team developed 13 medium-fidelity prototypes. These prototypes reflected different screens that Marillac Place staff and residents would have access to including a main/home page, a daily tasks page, a housing plan page, a warnings page and a boutique page.

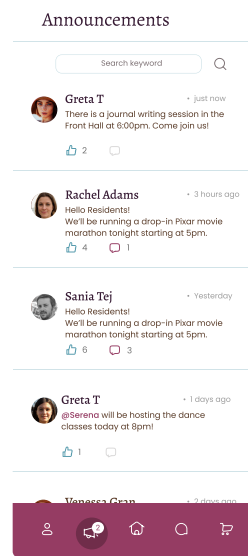
3.4.5 Follow-up Interview. We scheduled one more follow-up interview for Marillac Place staff to provide feedback on the updated prototypes. The second round of feedback was then incorporated into a higher-fidelity prototype. Some of the feedback from the Marillac Place staff included:

- (1) Screens to communicate between Marillac Place staff and residents (i.e. *Announcements* or *Chat* screens)
- (2) Summarizing tasks in a way that will be easier for residents to access and understand (i.e. *Today* and *Tasks* screens)

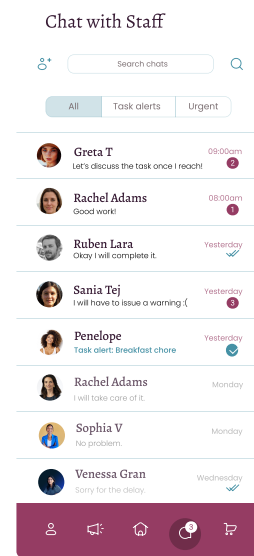
3.4.6 Higher-Fidelity Prototypes. Based on the follow-up interview, we developed high-fidelity prototypes (Figure 1) which incorporate key features designed to enhance user engagement and ensure a safe, supportive environment. Gameful elements such as badges and progress trackers are included to motivate residents and provide positive reinforcement by Marillac Place staff, which would help foster a sense of achievement and encouraging sustained participation in the PWAR program. We also included progress tracking which would nudge residents towards completing their daily tasks. We also included the ability for customization (e.g., resident tasks) to better support residents' independence, allowing individuals to personalize their experience and tailor the application to their unique preferences and needs. This embedded ability for customization also allows for the ease in updates to the general PWAR program, ensuring configurability should Marillac Place make modifications to their PWAR program, without having to hire developers to make changes to the code. Additionally, as raised by the Marillac Place team, this design prioritizes privacy and inclusivity by excluding features such as scoreboards, rankings, and in-app interactions between residents. These exclusions were intentional to prevent stress or conflict among Marillac Place residents and to safeguard resident confidentiality, ensuring the application remains a non-competitive and safe space.



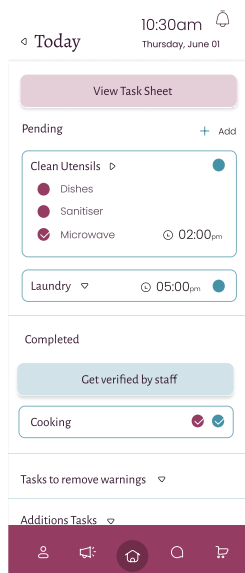
(a) Home provides a summary of daily tasks and announcements.



(b) Announcements from Marillac Place staff for residents.



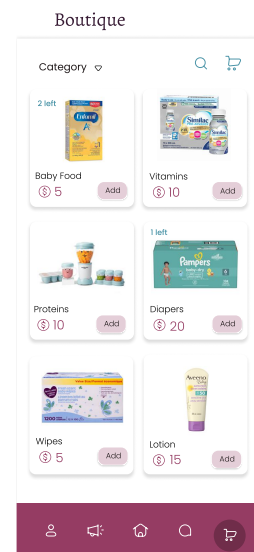
(c) Chat enables communication between residents and staff.



(d) Today provides a more detailed summary of each resident's daily tasks.



(e) Tasks provides a house-wide summary of tasks and assigned residents.



(f) Boutique enables residents to browse available rewards for completing tasks.

Figure 1: We developed prototypes for a Praise with a Raise app through a series of interviews and focus groups with Marillac Place staff. The app supports key tasks for residents including household tasks, communication with staff, and collecting rewards.

4 Discussion

We approached this project with the understanding that it be grounded in collaboration, trust, and meaningful engagement with the Marillac Place team and residents. This relationship helped us to recognize the strengths and unique perspectives brought forth from the participants and created unique co-learning opportunities. That is, we sought to conduct research *with*, not *on*, the community [18].

Curiosity has also been recognized as a key component of community-based research [6]. We relied on residents and staff to teach us about the Marillac Place community to improve our understanding of their needs and requirements. We asked the Marillac Place team about the day-to-day experiences and interactions of the team and residents. This process was highly iterative, with new information often leading to more meetings and questions, in order to co-design the application mockups. Though this process was time-intensive, it created multiple opportunities to build trust and a deeper understanding of requirements, to generate new ideas, and to gather feedback that grounded our design work in their real-life use cases.

4.1 Interdisciplinary Collaboration

Marillac Place approached the research team due to their interest in creating an innovative digital health app. We have since worked to build an inter-disciplinary, collaborative team to support Marillac Place through human-centred and client-centred project design. Our team has expertise in multiple disciplines, including user experience (UX) design, psychology, HCI and social work, all with research interests in developing novel and transformative health technologies.

We have leveraged this expertise in developing our research plan, methods, and process. For example, our discussions on human-centred design and client-centred approaches led us to include iterative, multi-phasic methods of collecting and incorporating data from stakeholders and end users of the app (i.e., Marillac Place residents and staff). Additionally, our conversations on persuasive system designs, the psychology of behaviour modification, and gamification led to the idea of incorporating digital gameful elements into the research, rather than simply creating a digitized version of the existing PWAR program. These fruitful discussions were only possible due to the interdisciplinary expertise of every project team member and our highly collaborative partnership with Marillac Place and have laid the foundation for a strong and successful collaboration moving forward.

4.2 Ongoing Changes to Requirements

One of the biggest challenges we face as external researchers is understanding how the existing PWAR program supports and impacts residents' lives. It was evident in our discussions with the Marillac Place team that we both approached design from a human-centred perspective, however their need to run a functional home vastly outweighs our scientific and design-based research questions. For example, during our requirements gathering and design activities the Marillac Place team made multiple changes to the structure of the program to reduce demands on residents to support their autonomy. The new program asks residents to only complete three core tasks each day and residents can customize additional tasks that are aligned with their personal goals. These changes were

made to increase participants' intrinsic motivation by building on their sense of autonomy [15].

In supporting Marillac Place, and understanding that the PWAR program will likely experience additional changes and iterations as the program evolves, we were aware that the requirements for the application may be a shifting target. We were also asked by Marillac Place to implement ways to increase the customization of the application so that the application could also be used with future changes to the PWAR program. This required ongoing communication with Marillac Place as it was difficult for the research team to foresee the possible changes that may occur in PWAR's future. This ongoing communication helped support productive partnerships with the Marillac Place team.

4.3 Working within Real-World Constraints

An ongoing constraint when working with a small nonprofit organization like Marillac Place was the limited resources available for this project, which required us to be flexible. In short, community-based organizations do not typically have the resources or expertise required to develop novel software. Recognizing these challenges, we sought out multiple sources of funding (internal institutional competitions, MITACS and Social Sciences and Humanities Research Council). We also reached out to an established student club, BluePrint, that focuses on building technology for social good, to find a team who can continue to co-design, develop and support ongoing maintenance of the application after the research project has concluded. Not only will this partnership with BluePrint provide the Marillac Place team with ongoing support after the application has been created, but it will also provide students interested in HCI to gain experience working with community-based organizations. Given these multiple layers of collaborations, and the difficulties in coordinating different resources like money, equipment, and expertise, it becomes crucial to establish clear communication channels and identify shared goals. We found that flexibility in project timelines, creative problem-solving, and leveraging existing community networks have been vital strategies to overcome resource limitations.

4.4 Limitations of Academic Research in Supporting Community-Based Initiatives

A key limitation of this project lies in the disconnect between the academic process and the real-world implementation of community-based solutions. While our objective was to co-design, develop and *deploy* a gameful digital tool, we were unable to translate the co-design work into a fully developed app due to a lack of funding and resources. We then sought out support from BluePrint, an organization that develop apps for non-profit organizations, to address this gap. But ultimately they decided to design their own version development process, leaving much of our work unrealized.

Our experience highlights an urgent need for more robust funding mechanisms to support community-based projects beyond the conceptualization and design phases. This gap leaves community-based organizations to bear the burden of sustaining these initiatives, in this case, requiring them to seek additional resources to re-do existing work.

We call on funding bodies to reconsider their support for community-based research projects so that resources and funding can enable true partnerships that lead to development of tools for real-world use. This call serves as both a reflection and a call to action for funders to invest in opportunities that bridge the gap between academic innovation and community impact. By providing comprehensive and sustained funding, future initiatives aimed at addressing societal challenges can achieve their full potential, ensuring that valuable opportunities for real-world impact are not missed.

5 Conclusions

This work demonstrates that interdisciplinary, community-based collaboration can drive design of technology to address complex social challenges. Our work concretely identifies key design elements — such as personalized tasks, progress tracking, and secure data access — that directly address the operational challenges of the PWAR program. However, limitations in academic funding and resources prevented full-scale deployment, underscoring the need for sustained investment and robust partnerships.

As a call to action, this paper highlights the urgent need for funding bodies to support all stages of community-based projects, from conceptualization to deployment. By fostering true partnerships and providing the necessary resources, funders can help bridge the gap between academic innovation and real-world impact.

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A Interview Questions

- (1) Which parts of this design do you like?
- (2) Which parts of this design do you dislike?
- (3) What would make this design easier to use for you?
- (4) What was difficult for you when using this design?
- (5) what was easy for you when using this design?
- (6) In terms of building independence, does this design help you increase your independence? If yes, how so? If not, why is that and what can be changed to help improve independence?
- (7) In terms of learning new skills, does this design help you achieve learning new skills? If yes, how so? If not, why is that and what can be changed to help improve this?
- (8) In terms of being a more active community member, does this design help you engage more as a community member? If yes, how so? If not, why is that and what can be changed to help improve active engagement?
- (9) What would you like to add to this design?
- (10) What would you like less of in the next iteration of the app design?

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